22/01/2025
Morning

Aakash
Medical | IIT-JEE | Foundations

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2025 Phase-1
[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. From all the English alphabets, five letters are chosen and are arranged in alphabetical order. The total number of ways, in which the middle letter is 'M', is:

(1) 4356 (2) 6084

(3) 14950 (4) 5148

Answer (4)

Sol. _ _ M _ _

before M, after M

M is at 13th position

$\Rightarrow \left[{}^{12}C_2 \cdot (1)\right][1]\left[{}^{13}C_2(1)\right]$ ways

= 5148

2. Using the principal values of the inverse trigonometric functions, the sum of the maximum and the minimum values of $16((\sec^{-1}x)^2 + (\text{cosec}^{-1}x)^2)$ is:

(1) $18\pi^2$ (2) $31\pi^2$

(3) $24\pi^2$ (4) $22\pi^2$

Answer (4)

Sol. Let $f(x) = 16[(\sec^{-1}x)^2 + (\text{cosec}^{-1}x)^2]$

$$f(x) = 16\left[(\sec^{-1}x + \text{cosec}^{-1}x)^2 - 2(\sec^{-1}x)\left(\frac{\pi}{2} - \sec^{-1}x\right)\right]$$

$$= 16\left[\frac{\pi^2}{4} - \pi\sec^{-1}x + 2\sec^{-1}x\right], \sec^{-1}x \in [0,\pi] - \left\{\frac{\pi}{2}\right\}$$

$$= 16\left[2\left(\sec^{-1}x - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{4} - \frac{\pi^2}{8}\right]$$

$f(x)$ minimum at $\sec^{-1}x = \frac{\pi}{4}$ and maximum at $\sec^{-1}x = \pi$

$\Rightarrow f(x)_{min} = 2\pi^2$, $f(x)_{max} = 20\pi^2$

3. The number of non-empty equivalence relations on the set {1, 2, 3} is:

(1) 5 (2) 7

(3) 4 (4) 6

Answer (1)

Sol. Number of non-empty equivalence relations on the set {1, 2, 3} will be bell number 3 = 5

{{1,} {2}, {3}},

{{1, 2}, {3}}, {{1, 3}, {2}, {2, 3}, {1}},

{{1, 2, 3}}. Therefore, total non-empty equivalence relation is 5.

4. Let $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ be two lines. Then which of the following points lies on the line of the shortest distance between L_1 and L_2?

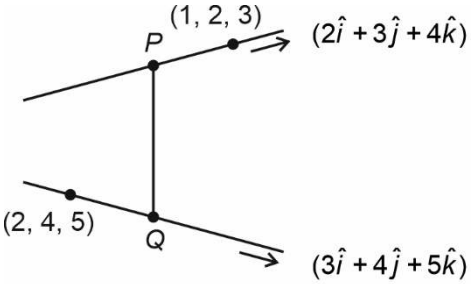
(1) $\left(\frac{8}{3}, -1, \frac{1}{3}\right)$ (2) $\left(2, 3, \frac{1}{3}\right)$

(3) $\left(\frac{14}{3}, -3, \frac{22}{3}\right)$ (4) $\left(-\frac{5}{3}, -7, 1\right)$

Answer (3)

Sol. $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$

$L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$





$P(2\lambda + 1, 3\lambda + 2, 4\lambda + 3)$

$Q(3\mu + 2, 4\mu + 4, 5\mu + 5)$

Dr's of $PQ < 2\lambda - 3\mu - 1, 3\lambda - 4\mu - 2,$

$4\lambda - 5\mu - 2 >$

$$PQ = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = -\hat{i} + 2\hat{j} - \hat{k}$$

$$\Rightarrow \quad \frac{2\lambda - 3\mu - 1}{-1} = \frac{3\lambda - 4\mu - 2}{2} = \frac{4\lambda - 5\mu - 2}{-1}$$

$$\Rightarrow \quad \lambda = \frac{1}{3} \quad \mu = \frac{-1}{6}$$

$$\Rightarrow \quad P\left(\frac{5}{3}, 3, \frac{13}{3}\right) \quad Q\left(\frac{3}{2}, \frac{10}{3}, \frac{25}{6}\right)$$

Dr's $PQ \langle 1, -2, 1 \rangle$

$\therefore$ Line

$$\frac{y - \frac{5}{3}}{1} = \frac{y - 3}{-2} = \frac{y - \frac{13}{3}}{1}$$

$\therefore$ Option (3) satisfies

5. Let the foci of a hyperbola be (1, 14) and (1, –12). If it passes through the point (1, 6), then the length of its latus-rectum is:

(1) $\frac{25}{6}$ (2) $\frac{288}{5}$

(3) $\frac{144}{5}$ (4) $\frac{24}{5}$

Answer (2)

Sol. $|SP - S'P| = 2a$

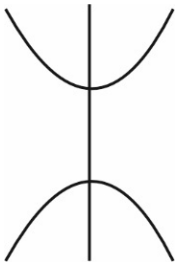
$SS' = 2ae$

$S(1, 14) \quad S'(1, -12) \quad P(1, 6)$

$2a = |8 - 18|$

$a = 5$

$2ae = 26 \Rightarrow ae = 13$



$$L(LR) = \frac{2b^2}{a} = \frac{2a^2(e^2 - 1)}{a}$$

$$= \frac{2[169 - 25]}{5} = \frac{288}{5}$$

6. A coin is tossed three times. Let X denote the number of times a tail follows a head. If μ and σ^2 denote the mean and variance of X, then the value of $64(\mu + \sigma^2)$ is:

(1) 48 (2) 32

(3) 64 (4) 51

Answer (1)

Sol.

	x_i	P_i
HHH	0	$\frac{1}{8}$
TTT	0	$\frac{1}{8}$
HHT	1	$\frac{1}{8}$
HTH	1	$\frac{1}{8}$
THH	0	$\frac{1}{8}$
TTH	0	$\frac{1}{8}$
THT	1	$\frac{1}{8}$
HTT	1	$\frac{1}{8}$



$$\mu = \sum x_i P_i = \frac{1}{2}$$

$$\sigma^2 = \sum x_i^2 P_i - \mu^2$$

$$= \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$64(\mu + \sigma^2) = 64\left[\frac{1}{2} + \frac{1}{4}\right]$$

$$= 64 \times \frac{3}{4} = 48$$

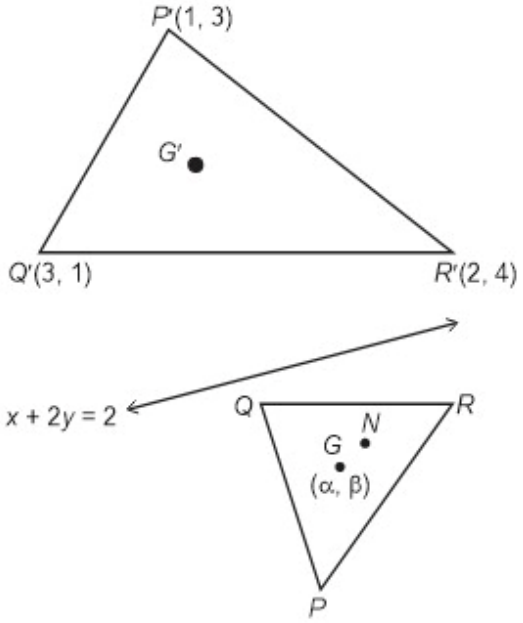
7. Let the triangle PQR be the image of the triangle with vertices (1, 3), (3, 1) and (2, 4) in the line $x + 2y = 2$. If the centroid of ΔPQR is the point (α, β), then $15(\alpha - \beta)$ is equal to:

(1) 19 (2) 24

(3) 22 (4) 21

Answer (3)

Sol. The centroid G'' (α, β) of ΔPQR be image of centroid of given triangle $P'Q'R'$.



Centroid of $\Delta P'Q'R'$ $= \left(\frac{1+3+2}{3}, \frac{3+1+4}{3}\right) = G'\left(2, \frac{8}{3}\right)$.

Image of $G'\left(2, \frac{8}{3}\right)$, w.r.t. line $x + 2y = 2$ is (α, β)

Then $\frac{\alpha - 2}{1} = \frac{\beta - \frac{8}{3}}{2} = \frac{-2\left(2 + \frac{16}{3} - 2\right)}{1+4}$

$\therefore \quad \frac{\alpha - 2}{1} = \frac{\beta - \frac{8}{3}}{2} = -\frac{32}{15}$

$\therefore \quad \alpha = -\frac{2}{15}$ and $\beta = -\frac{8}{5}$

Then $15(\alpha - \beta) = 15\left(-\frac{2}{15} + \frac{24}{15}\right) = 22$

8. Let $f(x)$ be a real differentiable function such that $f(0) = 1$ and $f(x + y) = f(x)\, f'(y) + f'(x)\, f(y)$ for all $x, y \in R$. Then $\sum_{n=1}^{100} \log_e f(n)$ is equal to:

(1) 2406 (2) 2384

(3) 2525 (4) 5220

Answer (3)

Sol. $\because f(x + y) = f(x).\, f'(y) + f'(x).f(y), \forall x, y \in R$ ...(i)

And $f(0) = 1$ ...(ii)

Now replace x by zero and y by zero we get

$f(0) = f(0)\, f'(0) + f'(0)\, f(0)$

$1 = f'(0) + f'(0)$

$\therefore \quad f'(0) = \frac{1}{2}$ ...(iii)

Now replace y by zero in equation (i), we get

$$f(x) = \frac{1}{2} f(x) + f'(x)$$

or, $\frac{1}{2} f(x) = f'(x)$

then $\frac{f'(x)}{f(x)} = \frac{1}{2}$

hence $\ln |f(x)| = \frac{x}{2} + c$

Put $x = 0$, we get $c = 0$

$\therefore \quad \ln |f(x)| = \frac{x}{2}$

Then $\sum_{n=1}^{100} \ln(f(\eta)) = \left(\frac{1}{2} + \frac{2}{2} + \frac{3}{2} + \ldots + \frac{100}{2}\right)$

$= \frac{5050}{2} = 2525$



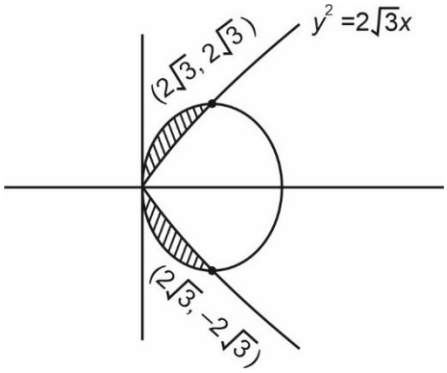
9. The area of the region, inside the circle $\left(x-2\sqrt{3}\right)^2+y^2=12$ and outside the parabola $y^2=2\sqrt{3}x$ is:

(1) $3\pi-8$ (2) $3\pi+8$

(3) $6\pi-16$ (4) $6\pi-8$

Answer (3)

Sol.



Required area $= 2\int_0^{2\sqrt{3}} \left(\sqrt{4\sqrt{3}x-x^2}-\sqrt{2\sqrt{3}x}\right)dx$

$= 2\int_0^{2\sqrt{3}} \left(\sqrt{12-\left(x-2\sqrt{3}\right)^2}-\sqrt{2\sqrt{3}x}\right)dx$

$= 2\left[\frac{x-2\sqrt{3}}{2}\sqrt{12-\left(x-2\sqrt{3}\right)^2}+\frac{12}{2}\sin^{-1}\left(\frac{x-2\sqrt{3}}{2\sqrt{3}}\right)-\frac{\sqrt{2\sqrt{3}}x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^{2\sqrt{3}}$

$= 2\{3\pi-8\}$

$= 6\pi-16$ sq. units.

10. Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms. If $a_1a_5 = 28$ and $a_2 + a_4 = 29$, then a_6 is equal to:

(1) 628 (2) 812

(3) 526 (4) 784

Answer (4)

Sol. $a_1a_5 = 28 \Rightarrow a^2r^4 = 28$

$a_2 + a_4 = 29 \Rightarrow ar + ar^3 = 29$

ar and ar^3 are roots of

$k^2 - 29k + 28 = 0$

$\Rightarrow k = 1, k = 28$

$ar = 1, ar^3 = 28$

$r^2 = 28, a^2 = \frac{1}{28}$

$a_6 = ar^5 \Rightarrow a_6^2 = a^2r^{10} = (28)^4$

$\Rightarrow a_6 = 784$

11. Let z_1, z_2 and z_3 be three complex numbers on the circle $|z| = 1$ with $\arg(z_1)=\frac{-\pi}{4}$, $\arg(z_2)=0$ and $\arg(z_3)=\frac{\pi}{4}$. If $\left|z_1\bar{z}_2+z_2\bar{z}_3+z_3\bar{z}_1\right|^2 = \alpha+\beta\sqrt{2}, \alpha, \beta \in Z$, then the value of $\alpha^2 + \beta^2$ is:

(1) 31 (2) 29

(3) 41 (4) 24

Answer (2)

Sol. $|z| = 1$

$\arg(z_1)=-\frac{\pi}{4}$, $\arg(z_2)=0$, $\arg(z_3)=\frac{\pi}{4}$

$z_1 = |1|e^{-\frac{\pi}{4}} = \frac{1}{\sqrt{2}}-\frac{i}{\sqrt{2}}$

$z_2 = 1 + 0i$

$z_3 = \frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}$

$z_1\bar{z}_2 = \frac{1-i}{\sqrt{2}}$

$z_2\bar{z}_3 = \frac{1-i}{\sqrt{2}}$

$z_3\bar{z}_1 = \frac{(1+i)^2}{2}$

$z_1\bar{z}_2+z_2\bar{z}_3+z_3\bar{z}_1 = \sqrt{2}+i\left(1-\sqrt{2}\right)$

$\left|z_1\bar{z}_2+z_2\bar{z}_3+z_3\bar{z}_1\right|^2 = 5-2\sqrt{2}$

$\alpha = 5, \beta = -2$

$\alpha^2 + \beta^2 = 29$



12. Let $f : R \to R$ be a twice differentiable function such that $f(x + y) = f(x) f(y)$ for all $x, y \in R$. If $f'(0) = 4a$ and f satisfies $f''(x) - 3af'(x) - f(x) = 0$, $a > 0$, then the area of the region $R = \{(x, y) \mid 0 \le y \le f(ax), 0 \le x \le 2\}$ is:

(1) $e^2 - 1$ (2) $e^4 - 1$

(3) $e^2 + 1$ (4) $e^4 + 1$

Answer (1)

Sol. $f(x + y) = f(x) f(y)$

$\Rightarrow$ Take $f(x) = k^x$

$k^x \log(k) = f'(x)$

$\log k = 4a \Rightarrow k = e^{4a}$

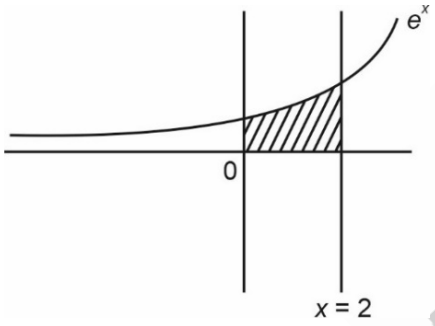
$f(x) = e^{4ax}$

$f''(x) - 3af'(x) - f(x) = 0$

$16a^2e^{4ax} - 12a^2e^{4ax} - e^{4ax} = 0$

$\Rightarrow \quad a^2 = \frac{1}{4} \Rightarrow a = \frac{1}{2}$

$f(x) = e^{2x}$



$\int_0^2 (e^x)dx = \boxed{e^2 - 1}$

13. Let $x = x(y)$ be the solution of the differential equation $y^2dx + \left(x - \frac{1}{y}\right)dy = 0$. If $x(1) = 1$, then $x\left(\frac{1}{2}\right)$ is

(1) $\frac{3}{2} + e$ (2) $3 - e$

(3) $3 + e$ (4) $\frac{1}{2} + e$

Answer (2)

Sol. $y^2dx + \left(x - \frac{1}{y}\right)dy = 0$

$y^2dx = \left(\frac{1}{y} - x\right)dy$

$\Rightarrow y^2 \frac{dx}{dy} = \frac{1}{y} - x$

$\Rightarrow \frac{dx}{dy} + \frac{x}{y^2} = \frac{1}{y^3}$

I.F. $= e^{\int \frac{1}{y^2}dy} = e^{\frac{-1}{y}}$

$\therefore$ Solution is

$xe^{\frac{-1}{y}} = \int e^{-\frac{1}{y}} \times \frac{1}{y^3}dy + C$

Let $\frac{-1}{y} = t$

$\Rightarrow \frac{1}{y^2}dy = dt$

$\Rightarrow xe^{-\frac{1}{y}} = -\int e^t t dt + C$

$\Rightarrow xe^{-\frac{1}{y}} = -e^t(t - 1) + C$

$\Rightarrow xe^{-\frac{1}{y}} = -e^{\frac{-1}{y}}\left(\frac{-1}{y} - 1\right) + C$

$x(1) = 1$

$\Rightarrow e^{-1} = -e^{-1}(-2) + C$

$\Rightarrow C = -e^{-1}$

$\Rightarrow x = \frac{1}{y} + 1 - e^{-1+\frac{1}{y}}$

$x\left(\frac{1}{2}\right) = 3 - e$



14. Two balls are selected at random one by one without replacement from a bag containing 4 white and 6 black balls. If the probability that the first selected ball is black, given that the second selected ball is also black, is $\frac{m}{n}$, where $gcd(m, n) = 1$, then $m + n$ is equal to

(1) 13 (2) 11

(3) 14 (4) 4

Answer (3)

Sol. Bag contains 4 white and 6 black balls

A : first ball selected is black

B : Second ball is also black

$$P\left(\frac{A}{B}\right) = \frac{\frac{6}{10}\times\frac{5}{9}}{\frac{4}{10}\times\frac{6}{9}+\frac{6}{10}\times\frac{5}{9}} = \frac{30}{24+30}$$

$$= \frac{30}{54} = \frac{5}{9}$$

$m + n = 5 + 9 = 14$

15. Let the parabola $y = x^2 + px - 3$, meet the coordinate axes at the points P, Q and R. If the circle C with centre at $(-1, -1)$ passes through the points P, Q and R, then the area of ΔPQR is

(1) 5 (2) 4

(3) 7 (4) 6

Answer (4)

Sol. $y = x^2 + px - 3$ is a parabola

at $x = 0 \Rightarrow y = -3$

Parabola cuts the y-axis at $(0, -3)$

Equation of circle will be

$(x + 1)^2 + (y + 1)^2 = (0 + 1)^2 + (-3 + 1)^2$

$\Rightarrow (x + 1)^2 + (y + 1)^2 = 5$

$\Rightarrow x^2 + y^2 + 2x + 2y - 3 = 0$

Now this circle cuts x-axis at $y = 0$

$x^2 + 2x - 3 = 0$

$\Rightarrow (x + 3)(x - 1) = 0$

$\Rightarrow x = -3, 1$

$(-3, 0)$ and $(1, 0)$

Points P, Q and R are $(0, -3)$ $(-3, 0)$ and $(1, 0)$

$$\text{Area of } (\Delta PQR) = \frac{1}{2}\begin{vmatrix} 0 & -3 & 1 \\ -3 & 0 & 1 \\ 1 & 0 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\times 12 = 6$$

$\Rightarrow$ Option (4) is correct.

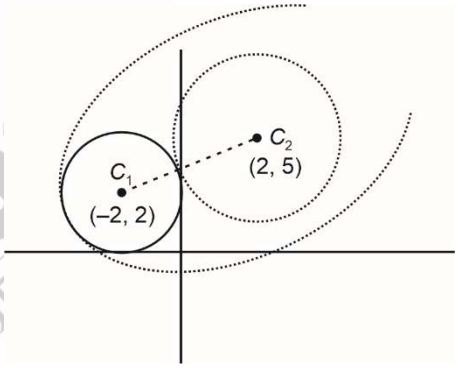
16. A circle C of radius 2 lies in the second quadrant and touches both the coordinate axes. Let r be the radius of a circle that has centre at the point $(2, 5)$ and intersects the circle C at exactly two points. If the set of all possible values of r is the interval (α, β), then $3\beta - 2\alpha$ is equal to:

(1) 12 (2) 14

(3) 15 (4) 10

Answer (3)

Sol.



$$C_1C_2 = \sqrt{(2+2)^2 + (5-2)^2}$$

$$= \sqrt{16+9}$$

$= 5$

$r + 2 > 5$

$r > 3$

$r < 5 + 2$

$r < 7$

$\therefore \alpha = 3, \beta = 7$

$3\beta - 2\alpha = 3(7) - 2(3) = 21 - 6 = 15$



17. If $\sum_{r=1}^{n} T_r = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$, then $\lim_{n\to\infty} \sum_{r=1}^{n}\left(\frac{1}{T_r}\right)$ is equal to

(1) 0 (2) 1

(3) $\frac{1}{3}$ (4) $\frac{2}{3}$

Answer (4)

Sol. $T_n = S_n - S_{n-1}$

$$= \frac{(2n-1)(2n+1)(2n+3)(2n+5) - (2n-3)(2n-1)(2n+1)(2n+3)}{64}$$

$$T_n = \frac{(2n-1)(2n+1)(2n+3)}{8}$$

$$\frac{1}{T_n} = 2\left(\frac{1}{(2n-1)(2n+1)} - \frac{1}{(2n-1)(2n+3)}\right)$$

$$S_n = 2\left(\frac{1}{1\times 3} - \frac{1}{(2n-1)(2n+3)}\right)$$

$$\lim_{n\to\infty} S_n = \frac{2}{3}$$

18. Let $A = \{1, 2, 3, \ldots, 10\}$ and $B = \left\{\frac{m}{n} : m, n \in A, m < n \text{ and } gcd(m, n)=1\right\}$. Then $n(B)$ is equal to

(1) 36 (2) 31

(3) 29 (4) 37

Answer (2)

Sol. Number of elements in set B, corresponding to

$m = 1$ are 9

$m = 2$ are $n = 3, 5, 7, 9 = 4$

$m = 3$ are $n = 4, 5, 7, 8, 10 = 5$

$m = 4$ are $n = 5, 7, 9 = 3$

$m = 5$ are $n = 6, 7, 8, 9 = 4$

$m = 6$ are $n = 7 = 1$

$m = 7$ are $n = 8, 9, 10 = 3$

$m = 8$ are $n = 9 = 1$

$m = 9$ are $n = 10 = 1$

Total number = 9 + 4 + 5 + 3 + 4 + 1 + 3 + 1 + 1 = 31

19. The product of all solutions of the equation $e^{5(\log_e x)^2+3} = x^8$, $x > 0$, is

(1) $e^{\frac{6}{5}}$ (2) e

(3) $e^{\frac{8}{5}}$ (4) e^2

Answer (3)

Sol. $e^{5(\log_e x)^2+3} = e^{\log x^8}$

$\Rightarrow 5(\log_e x)^2 + 3 = \log x^8 = 8\log_e x$

Put $\log_e x = t$, $5t(t-1) - 3(t-1) = 0$

$\Rightarrow 5t^2 - 8t + 3 = 0$

$\Rightarrow 5t^2 - 5t - 3t + 3 = 0$

$(5t-3)(t-1) = 0$

$t = 1 \Rightarrow \log_e x = 1 \Rightarrow x = e$

$t = \frac{3}{5} \Rightarrow \log_e x = \frac{3}{5} \Rightarrow x = e^{\frac{3}{5}}$

Product of all roots = $e^{1+\frac{3}{5}} = e^{\frac{8}{5}}$

Option = 3

20. Let for $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$, $I_1 = \int_0^{\frac{\pi}{4}} f(x)\,dx$ and $I_2 = \int_0^{\frac{\pi}{4}} x\, f(x)\,dx$. Then $7I_1 + 12I_2$ is equal to

(1) π (2) 2π

(3) 1 (4) 2

Answer (3)

Sol. $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$

$= 7\tan^6 x(1 + \tan^2 x) - 3\tan^2 x(1 + \tan^2 x)$

$= (7\tan^6 x - 3\tan^2 x)(1 + \tan^2 x)$

$= (7\tan^6 x - 3\tan^2 x)\sec^2 x$

$$I_1 = \int_0^{\frac{\pi}{4}} f(x)\,dx = \int_0^{\frac{\pi}{4}} (7\tan^6 x - 3\tan^2 x)\sec^2 x\,dx$$

$$= \left(\frac{7\tan^7 x}{7} - \frac{3\tan^3 x}{3}\right)\Big|_0^{\frac{\pi}{4}} = 1 - 1 = 0$$



$$I_2 = \int_0^{\frac{\pi}{4}} xf(x)\,dx = \int_0^{\frac{\pi}{4}} x(7\tan^6 x - 3\tan^2 x)\sec^2 x\,dx$$

$$= x(\tan^7 x - \tan^3 x)\Big|_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} 1.(\tan^7 x - \tan^3 x)\,dx$$

$$= 0 - \int_0^{\frac{\pi}{4}} \tan^3 x(\tan^2 x - 1)(\tan^2 x + 1)\,dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan^3 x - \tan^5 x)\sec^2 x\,dx = \frac{\tan^4 x}{4} - \frac{\tan^6 x}{6}\Big|_0^{\frac{\pi}{4}}$$

$$= \frac{1}{12}$$

Hence $7I_1 + 12I_2 = 1$

SECTION - B

Numerical Value Type Questions: This section contains 5Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. Let $\vec{c}$ be the projection vector of $\vec{b} = \lambda\hat{i} + 4\hat{k}, \lambda > 0$, on the vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$. If $|\vec{a} + \vec{c}| = 7$, then the area of the parallelogram formed by the vectors $\vec{b}$ and $\vec{c}$ is _____.

Answer (16)

Sol. $\vec{c} = \left(\frac{\vec{b}.\vec{a}}{|\vec{b}|}\right)\frac{\vec{a}}{|\vec{a}|}$

$$= \left(\frac{\lambda + 8}{9}\right)(\hat{i} + 2\hat{j} + 2\hat{k})$$

$$|\vec{a} + \vec{c}| = 7$$

$$\Rightarrow \left|\left(\frac{\lambda + 8}{9} + 1\right)\hat{i} + \left(\frac{2(\lambda + 8)}{9} + 2\right)\hat{j} + \left(\frac{2(\lambda + 8)}{9} + 2\right)\hat{k}\right| = 7$$

$$\left(\frac{\lambda + 8}{9} + 1\right)^2 + \left(\frac{2(\lambda + 8)}{9} + 2\right)^2 + \left(\frac{2(\lambda + 8)}{9} + 2\right)^2 = 49$$

$$\Rightarrow \lambda = 4 \Rightarrow \vec{c} = \frac{4}{3}\hat{i} + \frac{8}{3}\hat{j} + \frac{8}{3}\hat{k}$$

Area of parallelogram $= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{4}{3} & \frac{8}{3} & \frac{8}{3} \\ 4 & 0 & 4 \end{vmatrix} = 16$

22. If $\sum_{r=0}^{5} \frac{{}^{11}C_{2r+1}}{2r+2} = \frac{m}{n}$, $gcd(m, n) = 1$, then $m - n$ is equal to _____.

Answer (2035)

Sol. $(1+x)^{11} = {}^{11}C_0 + {}^{11}C_1 x + {}^{11}C_2 x^2 + \cdots + {}^{11}C_{11}x^{11}$

$$\int_0^1 (1+x)^{11}dx = \int_0^1 \left({}^{11}C_0 + {}^{11}C_1 x + {}^{11}C_2 x^2 + \cdots + {}^{11}C_{11}x^{11}\right)dx$$

$$\frac{(1-x)^{12}}{12}\Big]_0^1 = {}^{11}C_{0x} + \frac{{}^{11}C_1 x^2}{2} + \frac{{}^{11}C_2 x^3}{3} + \cdots + \frac{{}^{11}C_{11}x^{12}}{12}\Big]_0^1$$

$$\boxed{\frac{2^{12} - 1}{12} = C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \cdots + \frac{C_{11}}{12}} \quad \ldots(1)$$

Now,

$$\int_{-1}^0 (1+x)^{11}dx = \int_{-1}^0 \left({}^{11}C_0 + {}^{11}C_1 x + {}^{11}C_2 x^2 + \cdots + {}^{11}C_{11}x^{11}\right)dx$$

$$\frac{(1+x)^{12}}{12}\Big]_{-1}^0 = {}^{11}C_0 x + \frac{{}^{11}C_1 x^2}{2} + \frac{{}^{11}C_2 x^3}{3} + \cdots + \frac{{}^{11}C_{11}x^{12}}{12}\Big]_{-1}^0$$

$$\boxed{\frac{1}{12} = C_0 - \frac{C_1}{2} + \frac{C_2}{3}\cdots} \quad \ldots(2)$$

(1) – (2)

$$= \frac{2^{12} - 2}{12} = 2\left[\frac{C_1}{2} + \frac{C_3}{4} + \cdots\right]$$

$$\Rightarrow \sum_{r=0}^{5} \frac{C_{2r+1}}{2r+2} = \frac{2^{11} - 1}{12} = \frac{2047}{12} = \frac{m}{n}$$

$= 2047 - 12 = 2035$



23. Let the function,

$$f(x)=\begin{cases}-3ax^2-2, & x<1\\ a^2+bx, & x\geq 1\end{cases}$$

be differentiable for all $x \in R$, where $a>1$, $b \in \boldsymbol{R}$. If the area of the region enclosed by $y = f(x)$ and the line $y = -20$ is $\alpha+\beta\sqrt{3}$, $\alpha, \beta \in Z$, then the value of $\alpha + \beta$ is _____.

Answer (34)

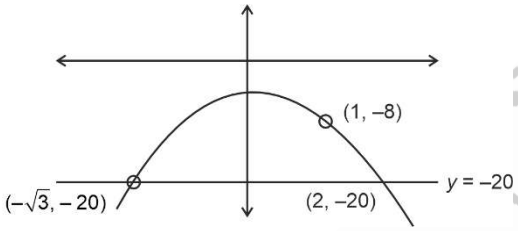
Sol. $f(x)$ is continuous and differentiable

at $x = 1$, LHL = RHL, LHD = RHD

$-3a - 2 = a^2 + b$, $-6a = b$

$a = 2$; $b = -12$

$$f(x)=\begin{cases}-6x^2-2, & x<1\\ 4-12x, & x\geq 1\end{cases}$$



$$\text{Area} = \int_{-\sqrt{3}}^{1}(-6x^2-2+20)\,dx+\int_{1}^{2}(4-12x+20)\,dx]$$

$$=16+12\sqrt{3}+6=22+12\sqrt{3}$$

$\therefore \quad \alpha + \beta = 34$

24. Let $L_1:\frac{x-1}{3}=\frac{y-1}{-1}=\frac{z+1}{0}$ and $L_2:\frac{x-2}{2}=\frac{y}{0}=\frac{z+4}{\alpha},\alpha\in R$, be two lines, which intersect at the point B. If P is the foot of perpendicular from the point $A(1, 1, -1)$ and L_2, then the value of $26\,\alpha(PB)^2$ is

Answer (216)

Sol. Point B

$(3\lambda+1,-\lambda+1,-1)\equiv(2\mu+2,\,0,\,\alpha\mu-4)$

$3\lambda + 1 = 2\mu + 2$

$-\lambda + 1 = 0$

$-1 = \alpha\,\mu - 4$

$\lambda = 1$, $\mu = 1$, $\alpha = 3$

$B\,(4, 0, -1)$

Let Point 'P' is $(2\delta + 2, 0, 3\delta - 4)$

Dr's of AP < $2\delta + 1, -1, 3\delta - 3$ >

$$AP \perp L_2 \Rightarrow \delta=\frac{7}{13}$$

$$P\left(\frac{40}{13},0,\frac{-31}{13}\right)$$

$$\therefore 26\alpha(PB)^2=26\times3\times\left(\frac{144}{169}+\frac{324}{169}\right)$$

$= 216$

25. Let A be a square matrix of order 3 such that $\det(A) = -2$ and $\det(3\text{adj}\,(-6\text{adj}\,(3A))) = 2^{m+n}\cdot3^{mn}$, $m > n$. Then $4m + 2n$ is equal to _____.

Answer (34)

Sol. As $A\,\text{adj}A = |A|I$, $\det(\lambda A) = \lambda^n \det A$

$\det(3\text{adj}(-6\text{adj}(3A))) = 3^3\det(\text{adj}(-6\text{adj}(3A)))$

$= 3^3(-6\text{adj}(3A))^2$

$= 3^3(-6)^6|3A|^4$

$= 3^9 2^6 \cdot 3^{12}\cdot(-2)^4$

$= 3^{21}\cdot 2^{10}$

Now comparing with given condition

$2^{m+n}\,3^{mn} = 2^{10}\cdot3^{21}$

$m + n = 10$, $mn = 21$

$\Rightarrow \quad m = 7$, $n = 3$ $(m>n)$

$\therefore \quad 4m + 2n = 28 + 6 = 34$



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. An electron is made to enter symmetrically between two parallel and equally but oppositely charged metal plates, each of 10 cm length. The electron emerge out of the electric field region with a horizontal component of velocity 10^6 m/s. If the magnitude of the electric field between the plates is 9.1 V/cm, then the vertical component of velocity of electron is

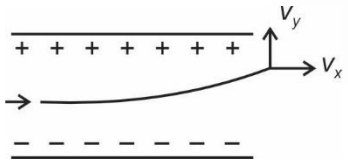
(mass of electron = 9.1×10^{-31} kg and charge of electron = 1.6×10^{-19} C)

(1) 0 (2) 16×10^6 m/s

(3) 16×10^4 m/s (4) 1×10^6 m/s

Answer (2)

Sol.



Horizontal component of velocity will remain same.

$$t = \frac{l}{v_x} = \frac{0.1}{10^6}$$

$$v_y = a_y t = \frac{eE}{m} t$$

$$= \frac{1.6\times10^{-19}\times9.1\times10^{2}}{9.1\times10^{-31}} \times \frac{0.1}{10^6}$$

$$= 1.6 \times 10^7$$

27. An amount of ice of mass 10^{-3} kg and temperature –10°C is transformed to vapour of temperature 110°C by applying heat. The total amount of work required for this conversion is,

(Take, specific heat of ice = 2100 $Jkg^{-1}K^{-1}$, specific heat of water 4180 $Jkg^{-1}K^{-1}$, specific heat of steam = 1920 $Jkg^{-1}K^{-1}$, Latent heat of ice = 3.35×10^5 Jkg^{-1} and Latent heat of steam = 2.25×10^6 Jkg^{-1})

(1) 3022 J

(2) 3043 J

(3) 3024 J

(4) 3003 J

Answer (2)

Sol. ice at –10°C ⟶ ice at 0°C ⟶ water at 0°C ↓ water at 100° ⟶ vapour at 100°C ⟶ vapour at 110°C

$$Q = 10^{-3}\left[2100\times10 + 3.35\times10^5 + 100\times4180 + 2.25\times10^6 + 10\times1920\right]$$

$$= 3043.2 \text{ J}$$

28. If B is magnetic field and μ_0 is permeability of free space, then the dimensions of (B/μ_0) is

(1) $ML^2T^{-2}A^{-1}$

(2) $L^{-1}A$

(3) $LT^{-2}A^{-1}$

(4) $MT^{-2}A^{-1}$

Answer (2)

Sol. For a current carrying loop at centre

$$B = \frac{\mu_0 i}{2R}$$

$$\therefore \quad \frac{B}{\mu_0} \equiv \frac{i}{R} \equiv [AL^{-1}]$$

